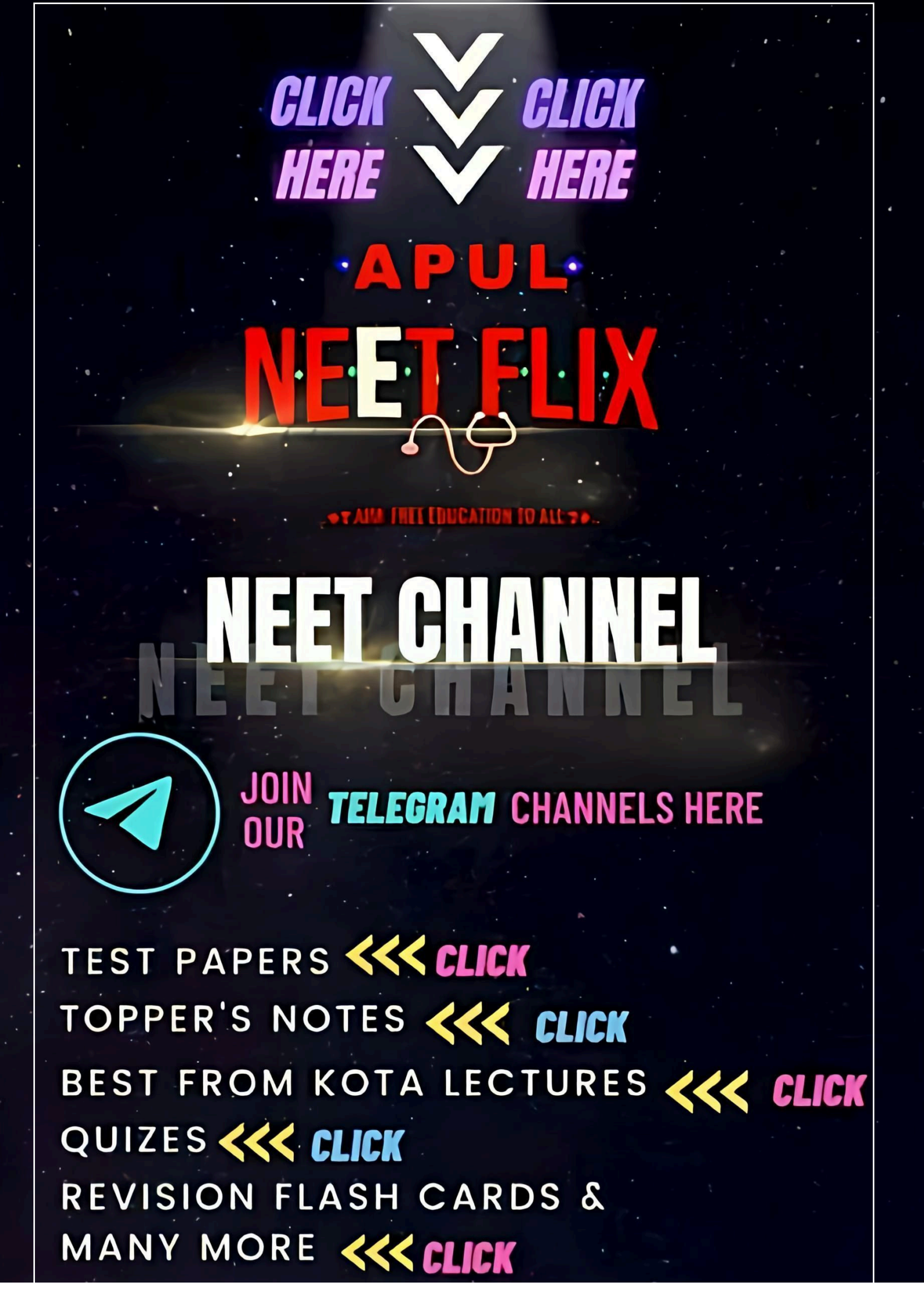
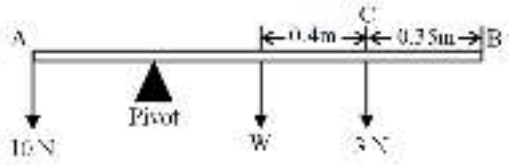

CLICK HERE
CLICK HERE
APUL
NEET FLIX
AIM FREE EDUCATION TO ALL
NEET CHANNEL
JOIN OUR TELEGRAM CHANNELS HERE
TEST PAPERS CLICK
TOPPER'S NOTES CLICK
BEST FROM KOTA LECTURES CLICK
QUIZES CLICK
REVISION FLASH CARDS &
MANY MORE CLICK

PHYSICS

1) A uniform rigid rod AB of length 1.5m and 200gm-wt is pivoted as shown. When a force of 10 N is applied at end A, and 3N at point C, calculate the distance of pivot point from end A when rod is in equilibrium. [g = 10^3 cm s^{-2}]



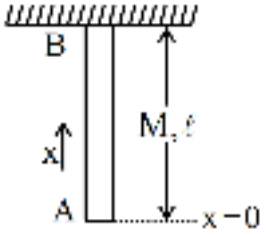
(1) 25 cm

(2) 33 cm

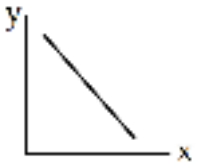
(3) 36 cm

(4) 42 cm

2) A uniformly dense rod with non-uniform young's modulus is hanging from ceiling under gravity of elastic potential energy density at every point is same, then which of the following graph correctly represent variation of young's modulus(Y) with distance x from lower end A to upper end B as shown in figure.



(1)



(2)



(3)



(4)



3) In experiment with resonance tube to find the speed of sound at room temperature and its end

correction, a resonance tube of length 150 cm is used first and second resonance are obtained when the length of air column in tube is 41 cm and 126 cm respectively with a tuning fork of frequency 200 Hz. Match the parameters in list-I with its suitable values in list-II.

	List-I		**List-II**
A	Wavelength of sound (in meter)	P	1.5
B	Minimum level of liquid in tube possible at resonance (cm)	Q	32
C	Speed of sound (ms^{-1})	R	340
D.	End correction (cm)	S	1.7
		T	24

(1) A→P, B→Q, C→T, D→S

(2) A→S, B→T, C→R, D→P

(3) A→T, B→S, C→R, D→P

(4) A→S, B→T, C→R, D→Q

4) To determine figure of merit of a galvanometer by half deflection method, a 5.0 V battery and a high resistance of 4900Ω are connected in series with galvanometer. Without any shunt, the galvanometer shows 20 divisions. When a shunt of 98Ω is connected, the deflection reduces to 10 divisions. The figure to merit of galvanometer is :

(1) 25 μA/division

(2) 40 μA/division

(3) 50 μA/division

(4) 80 μA/division

5) In experiment to determine refractive index of a glass slab using travelling microscope, following readings are obtained :

Microscope focused on	Reading
Cross mark without slab	1.215 cm
Cross mark with slab	2.115 cm
Top surface with lycopodium powder on it	4.115 cm

The refractive index of slab computed is :

(1) 1.40

(2) 1.45

(3) 1.50

(4) 1.55

6) Match the component given in column-II with its suitable behaviour mentioned in column-I, when it is connected with a bulb in series with a D.C supply.

	Column-I		Column-II
A	A bulb in series with it glows immediately and remains ON	1.	LED
B	Bulb glows for a moment then turns off	2.	Capacitor
C	Bulb takes time to reach full brightness	3.	Inductor
D	Bulb glows only in unidirection current flow but component does not emit any light	4.	Diode
E	Both bulb and component glows in unidirectional current only	5.	Resistor

(1) A→2, B→3, C→5, D→4, E→1

(2) A→5, B→2, C→1, D→4, E→3

(3) A→5, B→2, C→3, D→4, E→1

(4) A→3, B→2, C→5, D→1, E→4

7) An electric dipole with dipole moment $\vec{p} = (3\hat{i} + 4\hat{j}) \times 10^{-30}$ C-m is placed in an electric field $\vec{E} = 4000\hat{i}$(N/C). An external agent turns the dipole slowly until its electric dipole moment becomes $\left(-4\hat{i} + 3\hat{j}\right) \times 10^{-30}$ C-m. The work done by the external agent is equal to :–

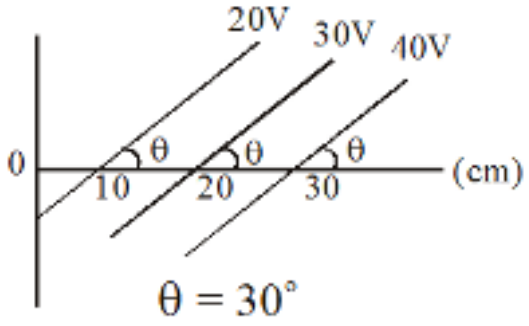
(1) 4×10^{-28} J

(2) -4×10^{-28} J

(3) 2.8×10^{-26} J

(4) -2.8×10^{-26} J

8) Some equipotential surfaces are shown in the figure. The magnitude and direction of the electric field is-



(1) 100 V/m making angle 120° with the x-axis

(2) 100 V/m making angle 60° with the x-axis

(3) 200 V/m making angle 120° with the x-axis

(4) None of these

9) An air capacitor of capacity 10 μF is connected to a constant voltage battery of 12V. Now the space between the plates is filled with a liquid of dielectric constant 5. The charge that flows now from battery to the capacitor is

(1) 120 μC

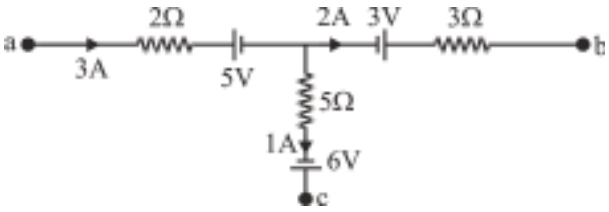
(2) 699 μC

(3) 480 μC

(4) 24 μC

10)

Find the potential difference (V_a – V_b) between a and b in the given circuit



(1) 14 V

(2) 17 V

(3) 12 V

(4) –14V

11) Two electric bulb, rated (25W, 220V) and (100W, 220V) are connected in series across a 220V voltage source. If the 25W and 100W bulb draw power P_1 and P_2 respectively then :

(1) P_1 = 9W, P_2 = 16W

(2) P_1 = 4W, P_2 = 16W

(3) P_1 = 16W, P_2 = 4W

(4) P_1 = 16W, P_2 = 9W

12) A solenoid has core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate H (magnetising field) and I (Intensity of magnetization).

(1) 2×10^3 Am^{-1}, 798×10^3 Am^{-1}

(2) 1.0 Am^{-1}, 1.5×10^5 Am^{-1}

(3) 6.8×10^{-5} Am^{-1}, 1.2×10^{-5} Am^{-1}

(4) 2.1×10^{-4} Am^{-1}, -2.6×10^5 Am^{-1}

13) A proton of mass 1.67×10^{-27} kg and charge 1.6×10^{-19} C is projected in xy plane with a speed of 2×10^6 m/s at an angle of 60° to the x-axis. If a uniform magnetic field of 0.104 T is applied along the y-axis, the path of the proton is :-

(1) A circle of radius 0.2 m and time period $\pi \times 10^{-7}$s

(2) A circle of radius 0.1 m and time period of $2\pi \times 10^{-7}$s

(3) A helix of radius 0.1 m and time period of $2\pi \times 10^{-7}$s

(4) A helix of radius of 0.2 m and time period $4\pi \times 10^{-7}$s

14) When a certain length of wire is turned into one circular loop, the magnetic induction at the centre of coil due to some current flowing is B_0. If the same wire is turned into three loops to make a circular coil, the magnetic induction at the center of this coil for the same current will be :-

(1) B_0

(2) $9B_0$

(3) $3B_0$

(4) $27B_0$

15) A stepdown transformer reduce the voltage of transmission line from 2200 V to 220 V. The power delivered by it is 880 W and its efficiency is 88%. The input current is :-

(1) 4.65 mA

(2) 0.465 A

(3) 0.4545 A

(4) 4.65 A

16) The values of L, C and R for a circuit are 1 H, 9 F and 3Ω. What is the quality factor for the circuit at resonance ?

(1) 1

(2) 9

(3) $\frac{1}{9}$

(4) $\frac{1}{3}$

17) A converging beam of ray is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens on the opposite side. If the lens is removed the point where the rays meet will move 5 cm closer to the lens. The focal length of the lens is :-

(1) –30 cm

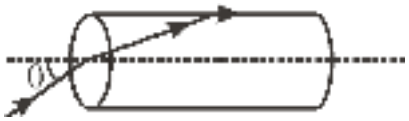
(2) 5 cm

(3) –10 cm

(4) 20 cm

18)

A transparent solid cylindrical rod has a refractive index of $2/\sqrt{3}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is

(1) $\sin^{-1}(1/2)$

(2) $\sin^{-1}(\sqrt{3}/2)$

(3) $\sin^{-1}(2/\sqrt{3})$

(4) $\sin^{-1}(1/\sqrt{3})$

19) In Y.D.S.E. the intensity at a point is $\left(\frac{1}{4}\right)$ of the maximum intensity. Angular position of this

point is :-

(1) $\sin^{-1}\left(\frac{\lambda}{d}\right)$

(2) $\sin^{-1}\left(\frac{\lambda}{2d}\right)$

(3) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$

(4) $\sin^{-1}\left(\frac{\lambda}{4d}\right)$

20) If light of wavelength 4000 Å falls on a metal which has a stopping potential 1.4 volt against photoelectric emission, then what is the work function of the metal [Take $h = 6.6 \times 10^{-34}$ Js and $c = 3 \times 10^{8}$ ms^{-1}] :

(1) 1.7 eV

(2) 3.7 eV

(3) 5.7 eV

(4) 7.7 eV

21) Calculate the de Broglie wavelength of electron of kinetic energy 200 eV (mass = 1×10^{-30} kg, charge = 1.6×10^{-19} C, planks constant $h = 6.6 \times 10^{-34}$ Js) :-

(1) 9.60×10^{-11} m

(2) 8.25×10^{-11} m

(3) 6.25×10^{-11} m

(4) 5.00×10^{-11} m

22) Assuming that 200 MeV of energy is released per fission of ${}_{92}U^{235}$ atom. Find the number of fission per second required to release 2 kW power :-

(1) 3.125×10^{13}

(2) 6.25×10^{13}

(3) 3.125×10^{15}

(4) 6.25×10^{14}

23) The angular momentum of an electron in the hydrogen atom is $\frac{3h}{2\pi}$. Its kinetic energy will be :-

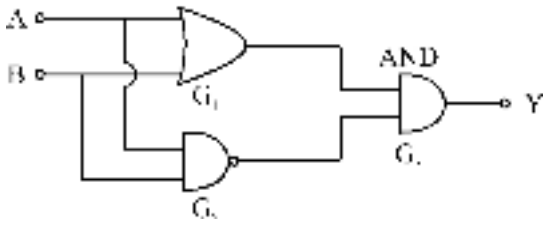
(1) 4.35 eV

(2) 1.51 eV

(3) 3.4 eV

(4) 6.8 eV

24) The following configuration of gate is equivalent to :



(1) NAND
(2) XOR
(3) OR
(4) None of these

25) ***Assertion (A) :-*** In a given situation of arrangement of charges, an additional charge is placed outside the Gaussian surface. In the Gauss' theorem $\oint \vec{E}.\,d\vec{s} = \frac{Q_{in}}{\epsilon_0}$, Q_{in} remains unchanged whereas electric field $\vec{E}$ at the site of the element is changed.
Reason (R) :- Electric field $\vec{E}$ at any point on the Gaussian surface is only due to inside charge.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

26) **Assertion (A) :** A bird sitting on a high power line (say 11kV) is not affected practically.
Reason (R) : Because no current flows through birds body, the circuit being incomplete.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

27) The resultant $\vec{C}$ of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to one vector $\vec{A}$, also $|\vec{C}| = |\vec{A}|$, then the angle between vector $\vec{A}$ and vector $\vec{B}$ is :-

(1) $\frac{\pi}{4}$ rad
(2) $\frac{3\pi}{4}$ rad
(3) $\frac{5\pi}{4}$ rad
(4) $\frac{7\pi}{4}$ rad

28) The mass 'm' of the heaviest stone which can be displaced by flowing river water depends on the velocity 'v' of flow of water, density 'ρ' of water and acceleration due to gravity 'g'. If mass 'm' of the stone is directly proportional to n^{th} power of velocity of flow, then what is the value of 'n' :-

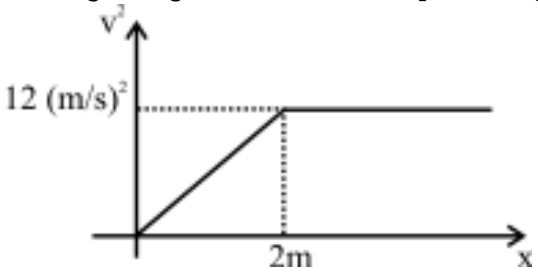
(1) 3

(2) 4

(3) 5

(4) 6

29) A particle moving along x-axis, its v^2 and position graph given in the figure. Initial acceleration of the object is:-



(1) zero

(2) 6 m/s^2

(3) 4 m/s^2

(4) 3 m/s^2

30) Two particles were thrown simultaneously from the same point one straight up, and the other at an angle of $\theta = 30°$ to the horizontal. The initial speed of each particle is equal to $v_0 = 25$ m/s neglecting air drag. Find the distance between the particles 1 sec after projection :-

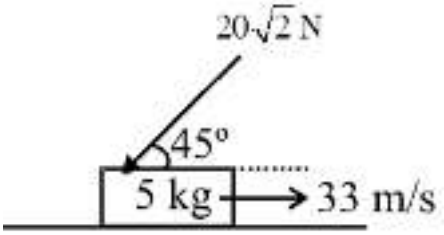
(1) $\frac{25\sqrt{3}m}{2}$

(2) 25m

(3) 12.5m

(4) 15m

31) A block of mass 5 kg is kept on a rough horizontal floor. It is given a velocity 33 m/s towards right. A force of $20\sqrt{2}$ N continuously acts on the block as shown in the figure. If the coefficient of friction between block and floor is 0.5 the velocity of block after 3 seconds is ($g = 10$ m/s^2)



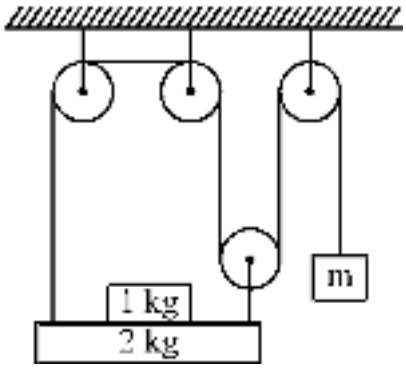
(1) 2 m/s

(2) Zero

(3) 33/12 m/s

(4) None of above

32) If the arrangement shown in the figure is in equilibrium then mass m of the block will be :-



(1) 3 kg

(2) 2 kg

(3) 1.5 kg

(4) 1 kg

33) A pump motor is used to deliver water at a certain rate from a given pipe. To obtain thrice as much water from the same pipe in the same time, power of the motor has to be increased to :

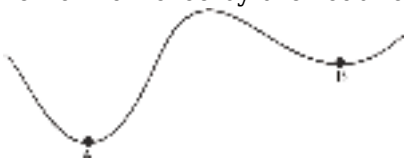
(1) 3 times

(2) 9 times

(3) 27 times

(4) 81 times

34) Car moves at a constant speed on a road as shown in figure. The normal force by the road on the car is N_A and N_B when it is at the points A and B respectively :



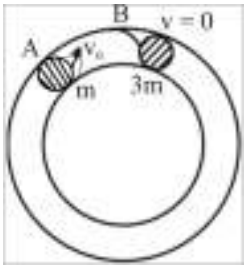
(1) $N_A = N_B$

(2) $N_A > N_B$

(3) $N_A < N_B$

(4) Insufficient information to decide the relation of N_A & N_B

35) In a smooth circular tube of radius R, a particle of mass 'm' moving with speed v_0 hits another particle of mass '3m' at rest as shown. The time after which the next collision takes place (assume headon elastic collision).



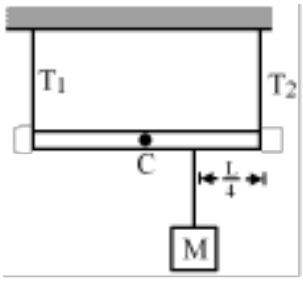
(1) $\frac{\pi R}{v_0}$

(2) $\frac{2\pi R}{v_0}$

(3) $\frac{\pi R}{2v_0}$

(4) $\frac{\pi R}{4v_0}$

36) Find tension T_1 and T_2 in the strings in the given diagram. Mass of the rod is M and length L



while mass of the block is also M.

(1) $T_1 = \frac{3Mg}{4}$, $T_2 = \frac{5Mg}{4}$

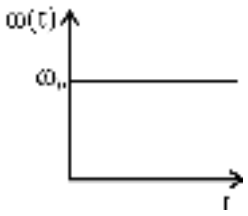
(2) $T_1 = \frac{5Mg}{4}$, $T_2 = \frac{3Mg}{4}$

(3) $T_1 = Mg$, $T_2 = Mg$

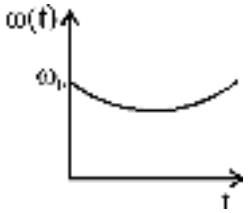
(4) $T_1 = \frac{3Mg}{2}$, $T_2 = \frac{Mg}{2}$

37) A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now the platform is given an angular velocity ω_0. When the tortoise move along a diametre of the platform with a constant velocity (with respect to the platform) the angular velocity of the platform $\omega(t)$ will vary with time t as :-

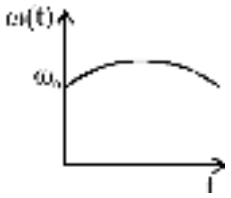
(1)

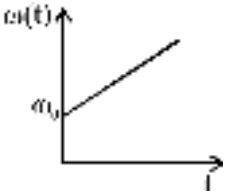


(2)



(3)



(4) 

38) A tank full of water has a small hole at its bottom. If three-fourth of the tank is emptied in t_1 second and the remaining one-fourth of the tank is emptied in t_2 second. Then, the ratio t_1/t_2 is :-

(1) $\sqrt{3}$
(2) 1
(3) $1/\sqrt{2}$
(4) $1/\sqrt{3}$

39) A ball rises to surface at a constant velocity in liquid whose density is 4 times than that of the material of the ball. The ratio of the force of friction acting on the rising ball and its weight is :

(1) 3 : 1
(2) 4 : 1
(3) 1 : 3
(4) 1 : 4

40) 1 cm^3 of water at 100°C and 1 atm pressure is heated to convert it to 1521 cm^3 vapour. Find increase in internal energy if latent heat of vaporization of water = 540 cal/g.

(1) 3.26 kJ
(2) 2.12 kJ
(3) 4.18 kJ
(4) 1.05 kJ

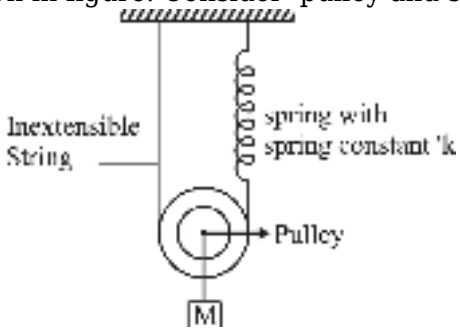
41) If E_m refers to the maximum intensity of radiations and λ_m is the wavelength corresponding to the maximum intensity of radiations, the Wein's displacement law refers to:-

(1) Increase in E_m when temperature is increasing
(2) Decreasee in E_m when temperature is increasing
(3) Increase in λ_m when temperature is increasing
(4) Decrease in λ_m when temperature is increasing

42) 10 gm of ice at –20°C is dropped into a calorimeter containing 10 gm of water at 10°C; the specific heat of water is twice that of ice. When equilibrium is reached, the calorimeter will contain :-

(1) 20 gm of water
(2) 20 gm of ice
(3) 10 gm ice and 10 gm of water
(4) 5 gm ice and 15 gm of water

43) Find the angular frequency ω (in rad/s) of mass M when displaced from its equilibrium position and then released for the system shown in figure. Consider pulley and string to be ideal.



[Given : M = 25 kg & k = 100 N/m]

(1) 1

(2) 1/2

(3) 4

(4) 2

44) A tuning fork produces 4 beats/s with another fork of frequency 288 cps. A little wax is placed on the unknown fork and it then produces 2 beats/s. The initial frequency of the unknown fork is :-

(1) 286 cps

(2) 292 cps

(3) 294 cps

(4) 288 cps

45) **Statement-1:** Two waves given by equations : $y_1(x, t) = 2a \sin(\omega t - kx)$ and $y_2(x, t) = a \sin(2\omega t - 2kx)$ will have equal intensity.
Statement-2: Intensity of waves propagating in same medium is proportional to square of amplitude only.

(1) Statement-1 is false, statement-2 is true

(2) Statement-1 is true, statement-2 is false

(3) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1

(4) Statement-1 is true, statement-2 is true; statement -2 is not correct explanation of statement-1.

CHEMISTRY

1) For copper atom match the column :

Column-I		**Column-II**	
(a)	Number of e^- with n = 4, ℓ = 0, m = 0	(i)	28
(b)	Number of e^- with n = 3, $s = +\frac{1}{2}$	(ii)	18
(c)	Number of e^- in penultimate shell	(iii)	9

(d)	Number of e^- in its kernel	(iv)	1

(1) (a)→(i), (b)→(ii), (c)→(iii), (d)→(iv)

(2) (a)→(iv), (b)→(iii), (c)→(i), (d)→(ii)

(3) (a)→(iv), (b)→(iii), (c)→(ii), (d)→(i)

(4) (a)→(iv), (b)→(ii), (c)→(iii), (d)→(i)

2)

The exothermic formation of ClF_3 is represented by the equation
$Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$; $\Delta H_r = -329$ kJ
Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2, F_2 and ClF_3 ?

(1) Removing Cl_2

(2) Increasing the temperature

(3) Adding inert gas at constant volume

(4) Decreasing the volume of the container

3) The incorrect statement for 14g of CO is :-

(1) It occupies 2.24 litre at NTP

(2) It corresponds to 0.5 mol of CO

(3) It corresponds to same mole as in 14 g of N_2

(4) It corresponds to 3.011×10^{23} molecules of CO

4)

How much of 80% pure $CaCO_3$ will be required to produce 44.8 L of CO_2 at STP?

(1) 200 g

(2) 100 g

(3) 180 g

(4) 250 g

5) For the reaction which is incorrect -

$LiH \rightarrow Li + \frac{1}{2} H_2$

(1) Oxidation state of lithium changes from +1 to zero.

(2) Oxidation state of hydrogen changes from –1 to zero.

(3) This is an example of disproportionation reaction.

(4) This is a redox reaction.

6)

Calculate the EMF of following cell
$Al_{(s)}|Al^{+3}(0.2M)||Fe^{+2}(0.01M)|Fe_{(s)}$

if $E^{o}_{Al/Al^{+3}}$ = 1.66 V and
$E^{o}_{Fe^{+2}/Fe}$ = – 0.44 V are given.

(1) 1.22 V
(2) 1.4 V
(3) 2.054 V
(4) 1.174 V

7)

If the half life of reaction increases as the initial concentration of substance increases, the order of reaction is

(1) Zero
(2) First
(3) Second
(4) Third

8) What will be volume of H_2 gas at NTP obtained by electrolysis of Aq. NaCl solution by passing 965 amp current for 10 second?

(1) 0.56 L
(2) 1.12 L
(3) 11.2 L
(4) 22.4 L

9) **Assertion (A):** The boiling point of aqueous solution of 0.1 M urea is less than that of 0.1 M KCl.
Reason (R): Elevation of boiling point is directly proportional to the number of moles of non-volatile solute particles present in the solution.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) is incorrect but (R) is correct.

10) A and B are two liquids that form an ideal solution. The vapour pressure of pure A is 40 mm of Hg at 310 K. The partial vapour pressure of liquid A in the solution of A and B was found to be 32 mm of Hg. The mole fraction of 'B' in the solution is found to be :-

(1) 0.8
(2) 0.5
(3) 0.2
(4) 0.4

11) For the reaction at 298 K
$2A + B \rightarrow C$
ΔH = 400 kJ mol^{-1} and ΔS = 0.2 kJ K^{-1} mol^{-1} At what temperature will the reaction become

spontaneous considering ΔH and ΔS to be constant over the temperature range ?

(1) T > 2000 K
(2) T = 2000 K
(3) T < 2000 K
(4) All temperatures

12) The bond energies of C – C, C = C, H – H and C – H bonds are 350, 600, 400 and 410 kJ per mole respectively. The heat of hydrogenation of ethylene (C_2H_4) is :-

(1) –170 kJ mol^{-1}
(2) –260 kJ mol^{-1}
(3) –400 kJ mol^{-1}
(4) –450 kJ mol^{-1}

13)

For a first order reaction the slope of arrhenius plot (ln K v/s 1/T) is -4.8×10^3 K then calculate the value of activation energy of reaction.

(1) 79.8 KJ mol^{-1}
(2) 159.6 KJ mol^{-1}
(3) 39.9 KJ mol^{-1}
(4) –79.8 KJ mol^{-1}

14) pH of 0.1 M solution of weak acid HA is 3. The value of K_a of this acid is :-

(1) 2×10^{-7}
(2) 10^{-5}
(3) 10^{-7}
(4) 10^{-6}

15) For the titration of weak acid and strong base, which one is best suitable indicator for indication of end point ?

(1) Phenolphthalein (pH : 8.2 to 10)
(2) Methyl red (4.2 to 6.3)
(3) Methyl orange (3.1 to 4.4)
(4) Bromothymol blue (6 to 7.6)

16) Which of the following statements are incorrect
(A) Nobelium is transuranic element
(B) Non-metal have low melting and boiling point except carbon and Boron
(C) Ga and Si are semimetal
(D) Samarium is actinoid

(1) A and B

(2) C and D

(3) Only D

(4) B and C

17) From the given compounds. How many compounds are acidic in nature.
Li_2O, SiO_2, Cl_2O_7, B_2O_3, Cr_2O_3, CO, CO_2, SO_2, MgO, V_2O_5

(1) 5

(2) 4

(3) 3

(4) 7

18) Match each of the diatomic molecules/ions in Column I with its property / properties in Column II. (give best suitable answer)

Column-I		**Column-II**	
(A)	B_2	(P)	Paramagnetic
(B)	N_2	(Q)	undergoes oxidation
(C)	O_2^-	(R)	Undergoes reduction
(D)	O_2	(S)	Bond order ≥ 2
		(T)	Mixing of 's' and `p' orbitals

(1) A→ P,Q,R,T : B → Q,R,S,T : C → P,Q,R : D → P,Q,R,S

(2) A→ P,Q,S,T : B → P,R,S,T : C → R,S,T : D → Q,R,S,T

(3) A→ Q,R,S,T : B → P,R,S,T : C → P,S,T : D → P,Q,R,T

(4) A→ P,Q,R,S : B → P,Q,R,S : C → P,Q,S : D → P,R,S,T

19)

Identify the correct orders against the property mentioned
(A) $H_2O > NH_3 > CHCl_3$ – dipole moment
(B) $XeF_4 > XeO_3 > XeF_2$ – number of lone pairs on central atom
(C) O–H > C–H > N–O – bond length
(D) $N_2 > O_2 > H_2$ – bond enthalpy
Choose the **correct** answer from the options given below :

(1) A, D only

(2) B, D only

(3) A, C only

(4) B, C only

20) Match the ion/compounds in column-I with its structure in column-II and select correct code -

	Column-I		**Column-II**
(a)	XeF_3^+	(i)	see saw

(b)	$XeF_5^{\ominus}$	(ii)	Pyramidal
(c)	XeO_2F_2	(iii)	T-shape
(d)	XeO_3	(iv)	Pentagonal planer

Code:

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(i)	(ii)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(i)	(ii)	(iii)	(iv)

(1) 1

(2) 2

(3) 3

(4) 4

21) Among the following compounds, which one has highest lattice energy ?

(1) LiCl

(2) MgO

(3) NaCl

(4) LiF

22) Select which is heteroleptic complex & show facial-meridional isomerism -

(1) $PtCl_2.2NH_3$

(2) $PtCl_4.4NH_3$

(3) $PtCl_2.4NH_3$

(4) $PtCl_4.3NH_3$

23) Match Column-I with Column-II and select the correct answer with respect to hybridisation using the codes given below:

Column-I (Complex)		**Column-II (Hybridisation)**	
(I)	$[PtCl_4]^{-2}$	(p)	dsp^2 hybridisation
(II)	$[Cu(CN)_4]^{3-}$	(q)	sp^3 hybridisation
(III)	$[Co(C_2O_4)_3]^{3-}$	(r)	sp^3d^2 hybridisation
(IV)	$[Fe(H_2O)_5NO]^{2+}$	(s)	d^2sp^3 hybridisation

Codes:

	(I)	(II)	(III)	(IV)
(1)	q	p	r	s
(2)	p	q	s	r
(3)	p	q	r	s
(4)	q	p	s	r

(1) 1

(2) 2

(3) 3

(4) 4

24)

Choose incorrect statements :
(i) CFT considered pure electrostatic attraction force between central metal ion and ligand.
(ii) $[Ti(H_2O)_6]^{3+}$ shows violet colour and this colour become more intense when it heated.
(iii) $[Co(NH_3)_6]^{3+}$ is more stable than $[CoCl_6]^{3-}$
(iv) More splitting will occur in presence of weak field ligand.

(1) (i) and (ii)

(2) (i) and (iii)

(3) (ii) and (iv)

(4) (iii) and (iv)

25) Which of the following statement is correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
I. a deep red vapor is evolved.
II. the vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
III. Chlorine gas is evolved
IV. chromyl chloride is formed

(1) I, II, IV

(2) I, II, III

(3) II, III, IV

(4) all are correct

26) Which of the following options is correct -

(1) Lanthanoid contraction is greater from element to element than actinoid contraction.

(2) Lanthanoids can achieve higher oxidation state more easily as compare to actinoid.

(3) Mn^{+3}, Er^{+3}, Ce^{+4} ions produces colourless aquated ion.

(4) On reaction with boiling water actinide gives a mixture of oxide & hydride.

27) Which of the following is correctly matched for d block elements :-

	Property	**Reason**

(1)	Catalytic	High reactivity
(2)	Alloy formation	Variable oxidation state
(3)	Formation of coloured ions	Due to excessive hydration
(4)	Nearly same size of 4d and 5d series elements	Effect of lanthanide contraction

(1) 1

(2) 2

(3) 3

(4) 4

28) Consider the following statements :
(A) I_2O_5 is a very good oxidising agent and is used in the estimation of carbon monoxide.
(B) ClO_2 is used as a bleaching agent.
(C) Decreasing order of stability of oxide formed by halogens is I > Br > Cl.
(D) Oxidising power of oxides of halogens is Cl = Br > I.
(E) Fluorine forms only one oxyacid while other halogens form a number of oxyacids.
Correct statements are :

(1) A, B and D only

(2) A, B, C, D only

(3) A, B and E only

(4) B, C, D and E only

29) On the basis of following experiment
Mixture + $K_2[Hg(I)_4]$ + KOH → Brown precipitated
identity the cation present in the mixture

(1) NH_4^+

(2) Cl^-

(3) Br^-

(4) Ag^+

30) **Statement-I :** For group 15 elements ionic radius is not increase regularly
Statement-II : On moving top to Bottom stability of -3 oxidation increases.

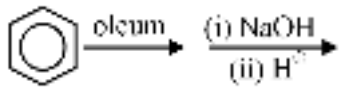
(1) **Statement I** and **Statement II** both are correct.

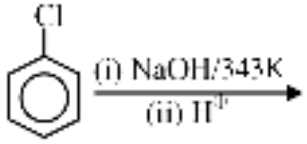
(2) **Statement I** and **Statement II** both are incorrect.

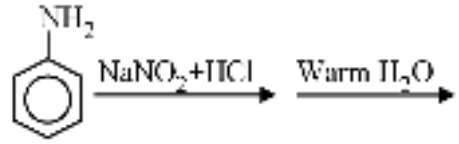
(3) **Statement I** is incorrect but **Statement II** is correct.

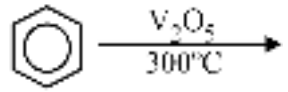
(4) **Statement I** is correct but **Statement II** is incorrect.

31) In which of the following reaction phenol is not obtained as major product

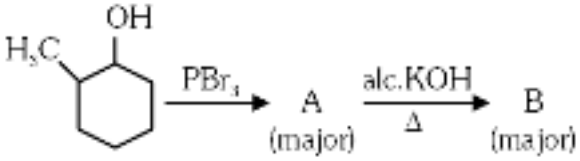
(1) 

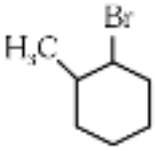
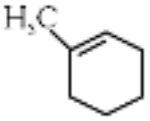
(2) 

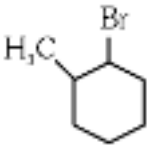
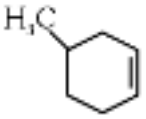
(3) 

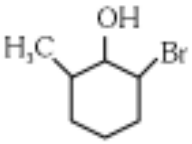
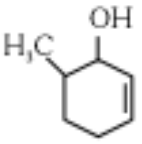
(4) 

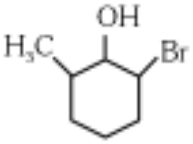
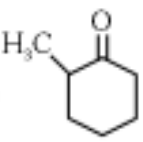
32) Major products A and B formed in the following reaction sequence, are



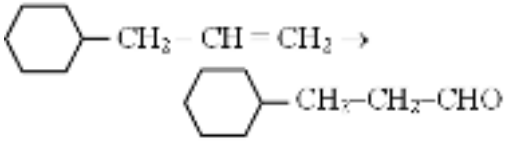
(1) A =  ; B = 

(2) A =  ; B = 

(3) A =  ; B = 

(4) A =  ; B = 

33) Identify the correct reagents that would bring about the following transformation



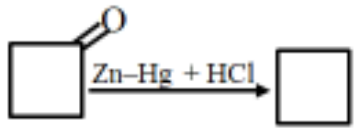
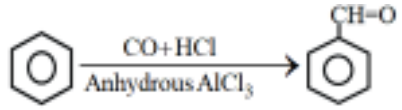
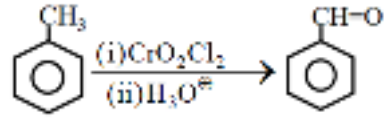
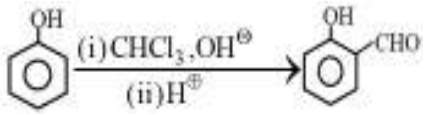
(1) (i) H_2O/H^+ (ii) CrO_3

(2) (i) BH_3 (ii) H_2O_2 / $\overset{\ominus}{O}H$ (iii) PCC

(3) (i) BH_3 (ii) H_2O_2 / $\overset{\ominus}{O}H$ (iii) Alk. $KMnO_4$ (iv) $H_3O^{\oplus}$

(4) (i) H_2O/H^+ (ii) PCC

34) Match the following columns-

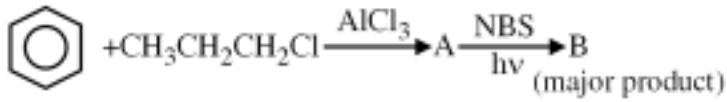
	Column-I (Reaction)		**Column-II (Name of reaction)**
A		P	Reimer-Tiemer Reaction
B		Q	Gatterman-Koch Reaction
C		R	Clemenson Reduction
D		S	Etard Reaction

(1) (A–R), (B–Q), (C–P), (D–S)

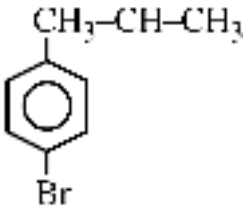
(2) (A–P), (B–Q), (C–R), (D–S)

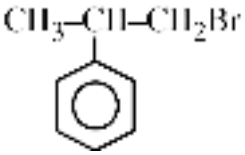
(3) (A–R), (B–Q), (C–S), (D–P)

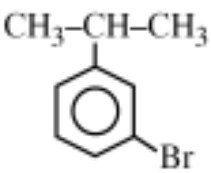
(4) (A–Q), (B–R), (C–S), (D–P)

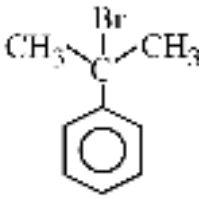
35) 

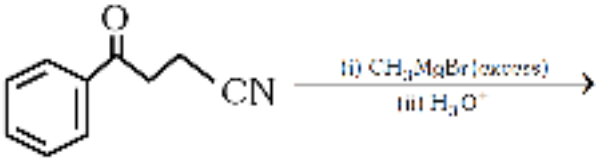
Identify product B.

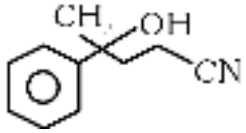
(1) 

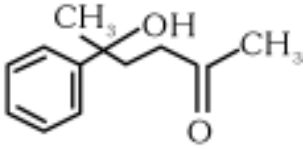
(2) 

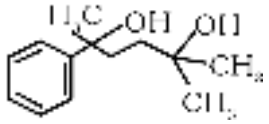
(3) 

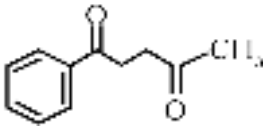
(4) 

36) The major product of the following reaction is 

(1) 

(2) 

(3) 

(4) 

37) **Assertion (A) :** Tropylium carbocation is more stable than Benzyl carbocation.
Reason (R) : Tropylium carbocation is aromatic in nature and forms 7 resonating structure.

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false

(4) Assertion (A) is false and Reason (R) is also false

38) $CH_3COOH \xrightarrow[\Delta]{NH_3} X \xrightarrow[\Delta]{PCl_5} Y,$ Y is :

(1) $CH_3-\underset{\underset{O}{\|}}{C}-Cl$

(2) CH_3CN

(3) CH_3CH_2CN

(4) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

39) When glucose reacts with bromine water the main product is

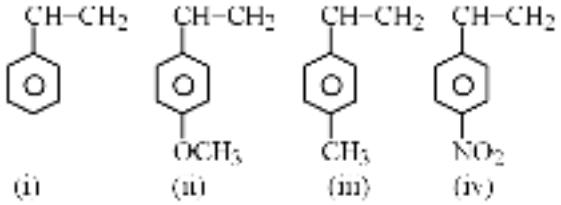
(1) Gluconic acid

(2) Glyceraldehyde

(3) Saccharic acid

(4) Acetic acid

40) Give decreasing order of reactivity towards electrophilic addition Reaction



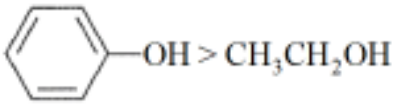
(1) (i) > (ii) > (iii) > (iv)

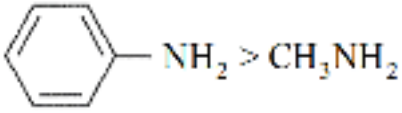
(2) (iv) > (iii) > (ii) > (i)

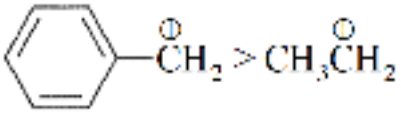
(3) (ii) > (i) > (iii) > (iv)

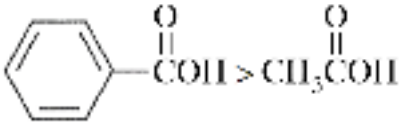
(4) (ii) > (iii) > (i) > (iv)

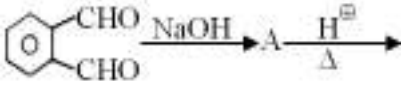
41) Which comparison is not correct as indicated ?

(1)  (acidic nature)

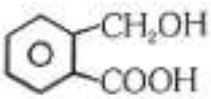
(2)  (basic nature)

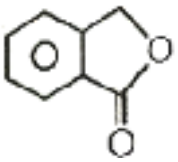
(3)  (stability)

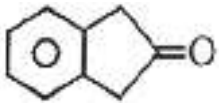
(4)  (acidic nature)

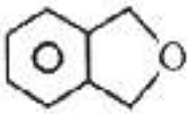
42)  B :

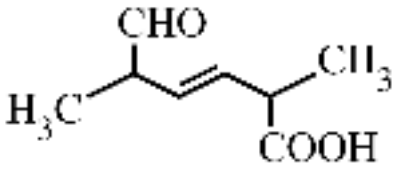
Major product B of reaction is :

(1) 

(2) 

(3) 

(4) 

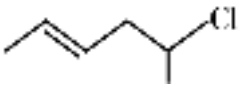
43) The IUPAC name for the following compound is :- 

(1) 2, 5-Dimethyl-6-carboxyhex-3-enal

(2) 6-Formyl-2-methylhex-3-enoic acid

(3) 2, 5-Dimethyl-5-carboxyhex-3-enal

(4) 2, 5-Dimethyl-6-oxohex-3-enoic acid

44) How many stereoisomer does this molecule have 

:-

(1) 2

(2) 4

(3) 6

(4) 8

45) Which of the following is false ?

(1) Steam distillation is applied to separate substances which are steam volatile and are immiscible with water.

(2) Distillation under reduced pressure is used to purify liquid having very high boiling point and decompose at or below their boiling point.

(3) Glycerol can not be separated from spent-lye in soap industry by distillation under reduce pressure.

(4) Simple distillation is used to separate volatile liquids from non volatile impurities.

BIOLOGY

1) Cohen and Boyer isolated an antibiotic resistance gene by cutting out a piece of DNA from a plasmid having antibiotic resistance in the year?

(1) 1963

(2) 1967

(3) 1972

(4) 1982

2) **Assertion (A)** :- The downstream processing and quality control testing vary from product to product.
Reason (R) :- The most commonly used bioreactors are of stirring type.

(1) A & R both are correct but R is not correct explanation of A

(2) A & R both are correct, and R is correct explanation of A

(3) A is correct but R is not correct

(4) A is incorrect but R is correct.

3) Match the column :

	Column-I		Column-II		Column-III
(a)	RNA interference	(i)	ADA deficiency	(t)	Kill insects
(b)	*Bacillus thuringiensis*	(ii)	Gene silencing mechanism	(u)	Hereditary disease
(c)	First clinical gene therapy	(iii)	Heat stable enzyme	(v)	DNA amplification
(d)	Taq polymerase	(iv)	Insecticidal protein production	(w)	Cellular defence

(1) a-ii-w, b-iii-t, c-iv-u, d-i-v

(2) a-iii-v, b-iv-w, c-ii-u, d-i-t

(3) a-ii-w, b-iv-t, c-i-u, d-iii-v

(4) a-ii-w, b-iv-t, c-iii-u, d-i-v

4) Match the Column-I with Column-II :-

	Column-I		Column-II
a	Micropropagation	i	To obtain virus free plants
b	Meristem culture	ii	Production of large number of plants
c	Biofortification	iii	Improvement of nutritional quality in crops
d	Somatic hybridization	iv	Protoplast fusion

(1) a-ii, b-i, c-iii, d-iv

(2) a-i, b-ii, c-iii, d-iv

(3) a-iii, b-i, c-ii, d-iv

(4) a-iv, b-iii, c-ii, d-i

5) Which of the following statement are incorrect ?
(a) RNAi takes place in all eukaryotic organisms as a method of cellular defence.
(b) C peptide along with A and B peptides are present in mature insulin.
(c) In 1990, the first transgenic cow, Rosie, produced human protein-enriched milk.
(d) Recombinant DNA technology serve the purpose of early diagnosis.
(e) 27 documented Basmati rice varieties are grown in India.

(1) b, c, d

(2) a, b, c

(3) a, d, e

(4) b and c only

6) Read the following statements :-
(A) Cyclosporin A is used as a clot buster.
(B) Whisky, brandy and rum are produced by distillation of fermented broth.
(C) NPV are the major baculoviruses used as biological control agents.
(D) The biogas production technology in India is developed by ICAR.
Which of the above statements are **correct**?

(1) A & B

(2) B & C

(3) C & D

(4) A & D

7) Match the column and choose the correct options.

	Column-I (Scientific Name)		**Column-II (Group)**		**Column-III (Feature)**
(A)	Sycon	(1)	Porifera	(i)	Pseudocoelomate parasite
(B)	Physalia	(2)	Cnidaria	(ii)	Bioluminescent
(C)	Pleurobrachia	(3)	Ctenophora	(iii)	Portuguese man-of-war
(D)	Taenia	(4)	Platyhelminthes	(iv)	Flame cells, endoparasite
(E)	Ascaris	(5)	Aschelminthes	(v)	Choanocytes (collar cells)

(1) A-1-v, B-2-ii, C-3-iii, D-4-iv, E-5-i

(2) A-1-i, B-2-ii, C-3-iii, D-4-iv, E-5-v

(3) A-1-v, B-2-iii, C-3-ii, D-4-iv, E-5-i

(4) A-1-iii, B-2-ii, C-3-i, D-4-iv, E-5-v

8) Identify the taxon of given animal.



(1) Cephalochordata
(2) Annelida
(3) Tunicata
(4) Hemichordata

9) Animal that usually undergoes external fertilisation with indirect development is–

(1) *Hyla*
(2) *Scoliodon*
(3) *Taenia*
(4) *Sycon*

10) **Statement-I :** The body cavity which is not lined by mesoderm is called coelom.
Statement-II : Those animals which do not have notochord are called non chordates.

(1) Statement-I & statement-II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I & statement-II are correct.

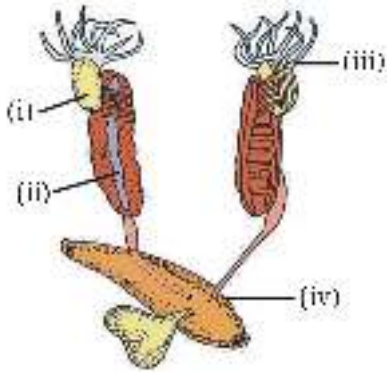
11) Cockroach sperm are stored in:-

(1) Phallic gland
(2) Long tubule
(3) Vas deferns
(4) Seminal vesicle

12) Which of the following sense organs of frog are not just cellular aggregation around nerve endings but are well organised

(1) sensory papillae
(2) taste buds and nasal epithelium
(3) eyes and internal ears
(4) none of these

13) Given below is the diagram of reproductive system of an animal labelled by (i), (ii), (iii) and (iv),



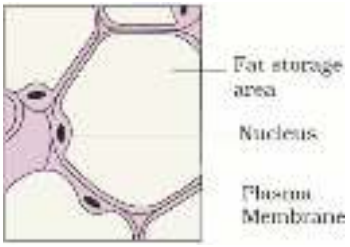
identify the animal with correct labelling of (i) to (iv)

(1) Male frog-(i) Testes (ii) Adrenal gland (iii) Fat bodies (iv) Cloaca
(2) Female frog-(i) Kidney (ii) Adrenal gland (iii) Ovary (iv) Cloaca
(3) Male frog-(i) Testes (ii) Bidder canal (iii) Adrenal gland (iv) Cloaca
(4) Female frog-(i) Ovary (ii) Oviduct (iii) Adrenal gland (iv) Cloaca

14) In cockroach, identify the parts of the foregut in **correct** sequence :

(1) Mouth → Oesophagus → Pharynx → Crop → Gizzard
(2) Mouth → Crop → Pharynx → Oesophagus → Gizzard
(3) Mouth → Gizzard → Crop → Pharynx → Oesophagus
(4) Mouth → Pharynx → Oesophagus → Crop → Gizzard

15) Identify the given below diagram and select the correct option ?



(1) Areolar tissue : It serves as a supportive frame form for epithelium
(2) Adipose tissue : It is another type of dense connective tissue located mainly beneath the skin.
(3) Areolar tissue : The cells of this tissue are specialised to store fats.
(4) Adipose tissue : it is another type of loose connective tissue located mainly beneath the skin.

16) **Assertion (A) :-** Birds and mammals have double circulation.
Reason (R) :- In birds and mammals, oxygenated and deoxygenated blood received by the left and right atria respectively passes on to the vertices of the same sides. The ventricles pump it out without any mixing. up i.e. two separate circulatory pathways are present in these organisms.

(1) A & R both correct, and R is correct explanation of A
(2) A & R both correct but R is not correct explanation of A
(3) A is correct but R is not correct
(4) A is incorrect but R is correct.

17) Which one of the following is not correct ?

(1) Normal activities of heart are regulated by specialised muscles tissues.
(2) Parasympathetic neural signals decreases the heart rate.
(3) Adrenal medullary hormones can also decreases the cardiac output.
(4) Human heart is myogenic & cardiac muscles are autoexcitable

18) The WBCs which makes maximum count amongst leucocytes is :-

(1) Monocytes – 60-65%
(2) Acidophils – 60-65%
(3) Basophils – 50-55%

(4) Neutrophils – 60-65%

19) Cardiac ouput is blood

(1) received by heart per minute
(2) pumped by ventricles per second
(3) pumped by left ventricle per minute
(4) pumped by left ventricle per hour

20) A chemosensitive area is highly sensitive to :-

(1) O_2 and H^+ ions
(2) CO_2 and Na^+ ions
(3) CO_2 and H^+ ions
(4) CO_2 and O_2

21) Match the column-I and II and choose correct option :-

	Column-I		**Column-II**
(P)	Carbaminohaemoglobin	(i)	70%
(Q)	Na^+ bicarbonate	(ii)	3%
(R)	Oxyhaemoglobin	(iii)	23%
(S)	O_2 dissolved in plasma	(iv)	97%

(1) P-i, Q-ii, R-iii, S-iv
(2) P-i, Q-iii, R-iv, S-ii
(3) P-iii, Q-i, R-iv, S-ii
(4) P-iii, Q-i, R-ii, S-iv

22) **Assertion-(A) :-** Glucose and amino acids, in the filtrate are 100% reabsorbed actively in the initial segment of nephron.
Reason-(R) :- Tubular epithelial cells of nephron in different segment perform reabsorption either by active or passive mechanism.

(1) (A) is true but (R) is false.
(2) (A) is false but (R) is true
(3) Both (A) and (R) are true and (R) is correct explanation of (A)
(4) Both (A) and (R) are true and (R) is not the correct explanation of (A)

23) Medullary gradient in kidney is mainly developed due to :

(1) NaCl and Urea
(2) NaCl and Glucose
(3) Glucose and Urea
(4) Ammonia and Glucose

24) **Statement-I :** Pons consists of fibre tracts that inter connect different regions of the brain.
Statement-II : Three major regions make up the brain stem : Mid brain, pons and medula oblongata.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

25) When a neuron is not conducting any impulse, i.e., in resting, the axonal membrane is comparatively more permeable to :

(1) K^+ ions

(2) Na^+ ions

(3) Ca^{2+} ions

(4) Equally permeable to both Na^+ and K^+ ions

26) Which one of the following is correct pairing of a body part and the kind of muscle tissue that moves it:

(1)	Heart wall	Involuntary unstriated muscle
(2)	Biceps of upper arm	Smooth muscle fibers
(3)	Nodal tissue	Smooth muscle
(4)	Iris	Involuntary smooth muscle

(1) 1

(2) 2

(3) 3

(4) 4

27) **Statement I :** Skeletal muscle are closely associated with the skeletal components of the body.
Statement II : Based on appearance, cardiac muscles are nonstriated.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

28) Wild contractions of muscles due to low Ca^{2+} in body fluid is called as:

(1) Arthritis

(2) Tetany

(3) Osteoporosis

(4) Gout

29) Due to insulin secretion:

(1) Increase in the level of ketone bodies in the blood
(2) Increase in the level of glucose in the blood
(3) Increase in level of glucose and ketone body in blood
(4) Decrease in the level of glucose in the blood

30) The hormone which acts through intracelluar receptor protein is–

(1) Growth hormone
(2) Prolactin
(3) Estrogen
(4) Adrenaline

31) Read all the given options carefully and select incorrect one-

(1) Pars nervosa synthesises two hormones
(2) PTH plays major role in Ca^{2+} homeostasis
(3) Thymosin plays major role in cell-mediated immunity
(4) Androgen influence libido

32) Given below are two statements :
Statement-I : Small amounts of androgenic steroids are also secreted by the adrenal cortex which play a role in the growth of axial hair, pubic hair and facial hair during puberty.
Statement-II : Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions and helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

33) Secretion of which gland destroys the acidic medium of male urethra before release of semen ?

(1) Seminal vesicle
(2) Cowpers gland
(3) prostate gland
(4) Pituitary gland

34) Ovulation in human a female takes place on 17th day of first menstrual bleeding. The duration of menstrual cycle in that female should be of:

(1) 05 Days
(2) 19 Days
(3) 28 Days

(4) 31 Days

35) In test tube baby programme, the embryo more than 8 blastomeres is transferred to uterus, this is called as:

(1) ZIFT
(2) GIFT
(3) IUT
(4) ICSI

36) **Assertion (A):** The testes are located outside the abdominal cavity in a pouch called the scrotum.
Reason (R): The scrotum maintains the testes at a temperature 2–2.5°C lower than the normal internal body temperature, which is essential for spermatogenesis.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is incorrect
(4) Both (A) and (R) are incorrect

37) During a routine prenatal check up, a doctor detects a heartbeat in a developing embryo using a stethoscope. Later at around 20 weeks, the mother reports feeling the first fetal movements and ultrasound confirms the presence of fine hair and separated eyelids. Based on this information which of the following combination of fetal development events is correct matched with their approximate time of occurrence ?

(1) Heartbeat → End of 2nd month
(2) First fetal movements → 24 weeks
(3) formation of digits → 5th month
(4) Eyelid separation and eye lash formation → End of 2nd trimester.

38) Which of the following statement is correct?

(1) The Medical Termination of Pregnancy (Amendment) Act, 2018 was enacted by the government of India to reduce the incidence of illegal abortion, maternal mortality and morbidity.
(2) Nearly 45 to 50 million MTPs are performed in a year all over the world which accounts to 1/4th of the total number of conceived pregnancies in a year.
(3) If the pregnancy has lasted for 12 weeks to 24 weeks, for MTP, opinion from two registered medical practitioners must be taken.
(4) Multiload 375 is a hormone releasing IUD

39) How many of the following are bacterial diseases ?
(i) Typhoid
(ii) Common cold
(iii) Pneumonia
(iv) Dysentery
(v) Malaria
(vi) Ascariasis

(vii) Alkaptonuria

(1) Two
(2) Three
(3) Four
(4) Five

40) **Statement-(I) :-** Both bone marrow and thymus provide micro environments for the development and maturation of T. lymphocytes.
Statement-(II) :- The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen.
Choose the correct statements :-

(1) Both are correct statements.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both are incorrect.

41) Opium is extracted from

(1) Atropa belladona
(2) Papaver somniferum
(3) Erythroxylum coca
(4) Cannabis sativa

42) Interferons are secreted by :

(1) Liver
(2) Spleen
(3) Virus-infected cells
(4) Lymphocytes

43) Universe was formed about how many years ago ?

(1) 20 billion
(2) 13.8 billion
(3) 20 million
(4) 65 million

44) Match the Column I with Column II and choose the correct option

	Column-I		**Column-II**
(A)	Australopithecines	(i)	Evolved 1.5 mya
(B)	Homo habilis	(ii)	Essentially ate fruit

(C)	Homo erectus	(iii)	Brain capacity was 1400cc
(D)	Neanderthal man	(iv)	Brain capacity was 650-800cc

	(A)	**(B)**	**(C)**	**(D)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(i)	(iii)	(iv)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

45) **Assertion (A) :** The wings of a butterfly and the wings of a bird are analogous structures.
Reason (R) : They have a common origin but perform different functions.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

46) Who gave biological concept of species :

(1) Aristotle

(2) Ernst Mayr

(3) John Ray

(4) Carolus Linnaeus

47) Which of the following statement is incorrect :-

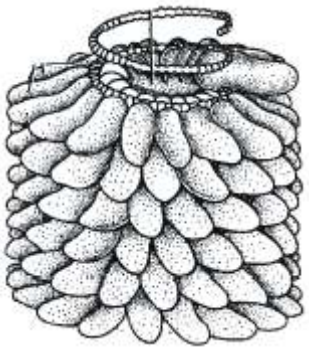
(1) Modern taxonomic studies based on only morphological character.

(2) The number and types of organism present on earth refers to biodiversity.

(3) 1.7 to 1.8 million species are known are described.

(4) Classification is the last step of taxonomic studies.

48) Which one of the following statement is incorrect about the plant virus seen here ?



(1) It is rod shaped

(2) It was discovered by Ivanowsky

(3) It was first crystallized by Stanley

(4) Its symmetry is icosahedral

49) In Carl Woese classification, how many kingdoms have unicellular organisms :-

(1) Two kingdom

(2) One kingdom

(3) Three kingdom

(4) Unicelluar organism are placed in domain not in kingdom

50) Study the following statement and give the answer:-
(A) Bacteria are sole members of the kingdom monera
(B) Bacteria are most abundant micro-organisms
(C) A handful of soil contain hundreds of bacteria
(D) In bacteria, mesosome help in respiration
(E) Bacteria show most extensive metabolic diversity
Select the most appropriate option from the following :

(1) A, B & E are correct C & D are incorrect

(2) A, C, E are correct B & D are incorrect

(3) A, B & C are correct D & E are incorrect

(4) A, B, C, D and E are correct

51) Which of the following alage contains floridean starch as reserve food material ?

(1) *Ectocarpus*

(2) *Gracilaria*

(3) *Volvox*

(4) *Fucus*

52) Match the column-I with column-II

Column-I	**Column-II**

(a)	Zoospores	(i)	Fusion between one large non motile female gamete and a smaller motile male gamete.
(b)	Isogamous	(ii)	Flagellated (Motile) Asexual spors
(c)	Anisogamous	(iii)	Gametes can be flagellated and similar in size
(d)	Oogamous	(iv)	Fusion of two gamete dissimilar in size

Choose the correct answer from the option given below :

	(a)	(b)	(c)	(d)
(1)	(ii)	(iii)	(iv)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(iv)	(iii)	(ii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

53) Given below are two statements :
Statement I : In a floral formula ⊕ stands for zygomorphic nature of the flower, and G stands for inferior ovary.
Statement II : In a floral formula ⊕ stands for actinomorphic nature of the flower and G stands for superior ovary.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

54) Asymmetric flower is found in.

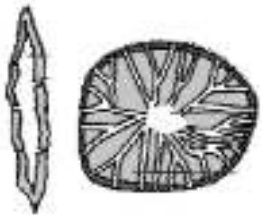
(1) *Sesbania*

(2) *Canna*

(3) Potato

(4) *Cassia*

55) Match the column I, II and III correctly :-

Column-I (Type)		**Column-II (Structure)**		**Column-III (Function)**	
(1)	Parenchyma	(a)		(i)	Mechanical strength and flexibility
(2)	Collenchyma	(b)		(ii)	Lignified cell provide fiber
(3)	Sclerenchyma	(c)		(iii)	Photosynthesis storage, secretion

(1) 1-a-iii, 2-b-i, 3-c-ii

(2) 1-a-ii, 2-b-i, 3-c-iii

(3) 1-a-i, 2-b-ii, 3-c-iii

(4) 1-a-iii, 2-b-ii, 3-c-i

56) The microtubule triplets making up centrioles are arranged in how many sets?

(1) 6 sets

(2) 9 sets

(3) 12 sets

(4) 18 sets

57) Match the column–A with column–B, and select the correct option from given below.

Column-A		**Column-B**	
a	RBC	i	Multinucleated
b	Fungi of phycomycetes	ii	Long and branched
c	Cork cells of bark	iii	Nucleus is absent
d	Neuron	iv	dead cells

(1) a - iii, b - i, c - iv, d - ii

(2) a - iv, b - iii, c - i, d - ii

(3) a - i, b - iii, c - ii, d - iv

(4) a - iv, b - iii, c - ii, d - i

58) **Assertion :** Every chromosome essentially has a primary constriction or the centromere.
Reason : The nuclear matrix or nucleoplasm contains nucleolus and chromatin.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

59) **Assertion (A) :-** Cell growth is a continuous process in terms of cytoplasmic increase.
Reason (R) :- DNA synthesis occurs only during two specific stages in the cell cycle.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) Both A and R are false.

60) Following are the events occur during cell cycle :
(A) Spliting of centromere
(B) Appearance of a furrow in the plasma membrane
(C) Doubling of amount of DNA in cell
(D) Most of the organelle duplication
(E) Arrangement of chromosomes at the equator of cell
(F) Condensation of chromatin to form chromosomes
Which of the following is the **correct sequence** of events ?

(1) A→B→C→D→E→F

(2) D→C→F→E→A→B

(3) F→E→D→C→B→A

(4) D→F→E→A→C→B

61) Arrange the following events of meiosis in correct order & select the correct option from options given below
(A) Crossing over
(B) Bivalent formation
(C) Centromere division
(D) Condensation of chromatin material initiates
(E) Chromosome condensation completes

(1) A, B, C, D, E

(2) B, A, D, E, C

(3) D, B, A, E, C

(4) D, B, A, C, E

62)

Name the class of enzyme that usually catalyze the following reaction :
$S - G + S^{\#} \rightarrow S + S^{\#} - G$
Where, G → a group other than hydrogen
S → a substrate
$S^{\#}$ → another substrate

(1) Hydrolase

(2) Lyase

(3) Transferase

(4) Ligase

63) Match the following:

Column-I		**Column-II**	
(A)	Pigment	(i)	Abrin
(B)	Lectins	(ii)	Anthocyanin
(C)	Drugs	(iii)	Vinblastin
(D)	Toxins	(iv)	Concanavalin A

(1) A–(iii), B–(i), C–(ii), D–(iv)

(2) A–(iv), B–(i), C–(ii), D–(iii)

(3) A–(iii), B–(i), C–(iv), D–(ii)

(4) A–(ii), B–(iv), C–(iii), D–(i)

64) Match List-I with List-II

	List-I		**List-II**
(A)	Primary CO_2 acceptor in C_3 cycle.	(i)	RuBP
(B)	Primary CO_2 acceptor in C_4 cycle.	(ii)	PEP
(C)	C_3 cycle	(iii)	Melvin Calvin
(D)	Law of limiting factor	(iv)	Blackmann

Choose the correct answer from the option given below-

(1) (A–i), (B–ii), (C–iii), (D–iv)

(2) (A–ii), (B–i), (C–iv), (D–iii)

(3) (A–i), (B–ii), (C–iv), (D–iii)

(4) (A–ii), (B–i), (C–iii), (D–iv)

65) Consider the given statements:
Statement I: In alcohol fermentation in yeast, pyruvic acid decarboxylase and alcohol dehydrogenase convert pyruvic acid to carbon dioxide and ethanol.
Statement II: In lactic acid fermentation, lactate dehydrogenase reduces pyruvic acid to lactic acid

and carbon dioxide in muscle cells during exercise.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

66) The complex II of mitochondrial electron transport chain is also known as

(1) Cytochrome bc_1 complex
(2) Succinate dehydrogenase complex
(3) Cytochrome c oxidase complex
(4) NADH dehydrogenase complex

67)

Which one is paired incorrectly ?

(1)	Auxin	–	Isolated from human urine & oat seedlings
(2)	Zeatin	–	Isolated from corn kernels
(3)	Ethylene	–	Gaseous hormone
(4)	Abscisic acid	–	Isolated from Fish sperm

(1) 1
(2) 2
(3) 3
(4) 4

68) Consider the given two statements:
Statement I: Spraying sugarcane crop with gibberellins increases the yield by as much as 20 tonnes per acre.
Statement II: Sugarcane stores carbohydrates as sugar in their stems and spraying sugarcane crop with gibberellins increases the length of the stem.

(1) Both Statement I and Statement II are correct and Statement II explains Statement I
(2) Both Statement I and Statement II are correct but Statement II does not explain Statement I
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

69) Egg apparatus of embryo sac is:

(1) 8 nucleate and 7 celled
(2) 8 cell and 7 nucellus
(3) having 3 cell
(4) present in pollen sac

70) How many meiosis are required for formaton of 800 male gametes.

(1) 200
(2) 400
(3) 800
(4) 100

71) Find out **incorrect** statement.

(1) The vegetative cell of pollen grain has large irregular shaped nucleus.
(2) Intine of pollen grain is made up of pectocellulose.
(3) Generative cell of pollen grain divides meiotically to give rise to the two male gametes.
(4) Each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.

72) When the population density reaches the carrying capacity that stage is called as :-

(1) Exponential phase
(2) Acceleration phase
(3) Deceleration phase
(4) Asymptote phase

73) Refer to the given table that summarizes the interactions between two organisms (organism 1 and organism 2). Identify the types of interaction (A, B and C) and select the correct answer.

		Effects on Organism 2		
		Benefit	**Harm**	**No Effect**
Effects on Organism 1	**Benefit**	Mutualism	Predation	**B**
	Harm	**A**	Competition	Amensalism
	No Effect	Commensalism	**C**	–

(i) A can be either predation or parasitism.
(ii) B can be either commensalism or amensalism.
(iii) C can be amensalism.
(iv) A can be amensalism.
Option :-

(1) (i) and (ii)
(2) (ii) and (iii)
(3) (iii) and (iv)
(4) (i) and (iii)

74) Observe the following table

Section-A	**Section-B**	**Interaction**
+	–	a
–	–	b
–	0	c

+	0	d

The correct order of a, b, c, d would be :

(1) a = Commensalism, b = Parasitism, c = Predation, d = Mutualism

(2) a = Predation, b = Competition, c = Amensalism, d = Commensalism

(3) a = Amensalism, b = predation, c = Commensalism, d = Parasitism

(4) a = Parasitism, b = Predation, c = Amensalism, d = Commensalism

75) Which of the following is correct regarding ecosystem ?

(1) Primary producer in aquatic ecosystem are species like zooplankton

(2) In terrestial ecosystem major producers are herbs & woody plants.

(3) Death of organism is end of detritus food chain

(4) Insect, birds & mammals in terrestial ecosystem are common decomposers.

76)

Match the table I, II and III.

I		**II**		**III**	
1	Tertiary consumer	a	Fourth trophic level	i	Birds, fishes, wolf
2	Secondary consumer	b	Third trophic level (primary carnivore)	ii	Man, Lion
3	Secondary Producer	c	First trophic level (plants)	iii	Phytoplankton, grass, tree
4	Primary producer	d	Second trophic level (herbivore)	iv	Zooplankton, grass hopper and cow

(1) 1-a-ii, 2-b-iii, 3-d-iv, 4-c-i

(2) 1-a-i, 2-b-ii, 3-c-iii, 4-d-iv

(3) 1-b-i, 2-a-ii, 3-d-iv, 4-c-iii

(4) 1-a-ii, 2-b-i, 3-d-iv, 4-c-iii

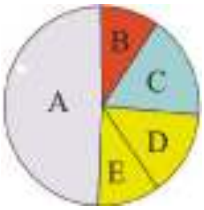
77) A dark coloured amorphous substance which is highly resistant to microbial activity is called :-

(1) Detritus

(2) Duff

(3) Litter

(4) Humus

78) 

Given figure shows representation of global biodiversity of vertebrates. Which alphabet represents "Reptiles" ?

(1) A
(2) B
(3) C
(4) D

79) Which one is not a "The evil Quartet" ?

(1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Co-evolution
(4) Alien species invasions

80) In which of the following sex is determined by female individual –

(1) Human
(2) Drosophila
(3) Birds
(4) Grasshopper

81) Which of the following is not a sex linked character?

(1) Haemophilia
(2) Colour blindness
(3) Hypertrichosis
(4) Phenylketonuria

82) Consider following genetic disorders-
Haemophilia, Sickle Cell Anaemia, Cystic fibrosis, Thalassemia, Phenyl Ketonuria, Down syndrome, Klinefelter's syndrome, Turner's syndrome
How many of the above disorder are Mendelian disorder?

(1) 5
(2) 2
(3) 8
(4) 3

83) Which of the following statement is correct?

(1) Morgan united knowledge of chromosomal segregation with Mendelian principles

(2) Thomas Hunt Morgan worked with *Drosophila melanogaster* to experimentally verify the chromosomal theory of inheritance

(3) A single mating of *Drosophila melanogaster* could produce a small number of progeny flies

(4) There was no clear differentiation of sexes in *Drosophila melanogaster*

84) Match List I with List II

	List I		**List II**
A.	Two or more alternative forms of a gene	I.	Back cross
B.	Cross of F_1 progeny with homozygous recessive parent	II.	Co-dominance
C.	Cross of F_1 progeny with any of the parents	III.	Allele
D.	AB blood group	IV.	Test cross

Choose the correct answer from the options given below :

(1) A–I, B–II, C–III, D–IV

(2) A–II, B–I, C–III, D–IV

(3) A–III, B–IV, C–I, D–II

(4) A–IV, B–III, C–II, D–I

85) The essential life processes such as metabolism, translations and splicing evolved around :-

(1) DNA

(2) Protein

(3) RNA

(4) Lipid

86) To prove that DNA is the genetic material, which radioactive isotopes were used by Hershey and Chase (1952) in their experiments?

(1) ^{35}S and ^{15}N

(2) ^{32}P and ^{35}S

(3) ^{32}P and ^{15}N

(4) ^{14}N and ^{15}N

87) The sequence of events mentioned below are symbolized by alphabets. Choose the correct answer where the alphabets are matched with the processes:

$$RNA \xrightarrow{(A)} DNA \xrightarrow{(B)} DNA \xrightarrow{(C)} \text{m-RNA} \xrightarrow{(D)} \text{Polypeptide}$$

(1) A-Replication, B-Transformation, C-Transcription, D-Translation

(2) A-Reverse transcription, B-Translation, C-Transcription, D-Replication

(3) A-Replication, B-Transcription, C-Translation, D-Tranduction

(4) A-Reverse transcription, B-Replication, C-Transcription, D-Translation

88) Read the following statements and choose the correct set of statements :
(a) The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope
(b) During transcription only a segment of DNA and both of the strands is copied into RNA
(c) In prokaryotes, the monocistronic structural genes have interrupted coding sequences
(d) Change in nucleic acids are responsible for change in amino acids in proteins
Choose the correct answer from the options given below :

(1) (a) and (c) only

(2) (a), (b), (c) and (d) only

(3) (a), (c) and (d) only

(4) (a) and (d) only

89)

Match the column-I with column-II and choose the correct answer :-

Column-I		**Column-II**	
(A)	Average bases in a human gene	(i)	Y-Chromosome
(B)	Dystrophin gene	(ii)	3000
(C)	Maximum genes	(iii)	2.4 million bases
(D)	Minimum genes	(iv)	Chromosome-1

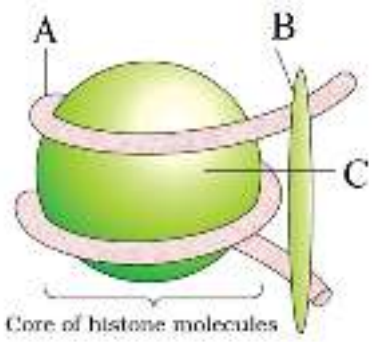
(1) A-(ii), B-(iv), C-(iii), D-(v)

(2) A-(ii), B-(iii), C-(iv), D-(i)

(3) A-(ii), B-(iv), C-(i), D-(iii)

(4) A-(iii), B-(iv), C-(ii), D-(i)

90) Refer the given figure of nucleosome and select the option that correctly identifies the parts of A, B and C



	A	B	C
(1)	DNA	Histone octamer	H1 histone
(2)	Histone octamer	H1 histone	H1 histone
(3)	Histone octamer	DNA	H1 histone
(4)	DNA	H1 histone	Histone octamer

(1) 1

(2) 2

(3) 3

(4) 4